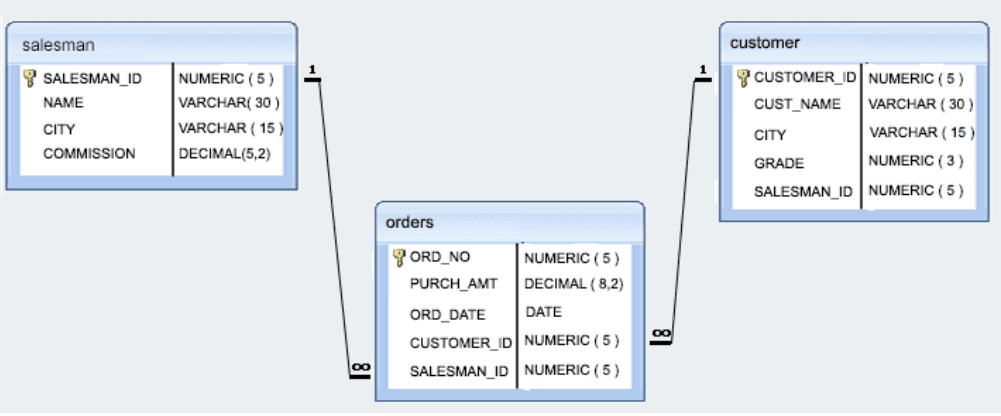


NIRMA UNIVERSITY
School of Technology, Institute of Technology
Master of Computer Application (2-Years Programme)
Semester-I

List of Practical

6CS104CC22	Database Management System	[3-0-2-4]
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Sr. No.	Title	Hours	CLO																				
1	Write and perform the SQL queries for the following: Create the below tables as specified:  <pre> salespeople(snum number(4) primary key,sname varchar2(20),city varchar2(15),comm number(5,2)); customer(cnum number(4) primary key,cname varchar2(20),city varchar2(15),rating number(4),snum number(4) references salespeople); orders(onum number(4) primary key,amt number(6,2),odate date,cnum number(4) references customer,snum number(4) references salespeople); </pre> Insert the records: SQL> insert into salespeople values(&snum,'&sname','&city',&comm); <table border="1"> <thead> <tr> <th><u>snum</u> number(4) primary key</th><th><u>sname</u> varchar2(20)</th><th><u>city</u> varchar2(15)</th><th><u>comm</u> number(5,2)</th></tr> </thead> <tbody> <tr> <td>101</td><td>Devendra</td><td>Ahmedabad</td><td>4.5</td></tr> <tr> <td>102</td><td>Yogini</td><td>Ahmedabad</td><td>25.52</td></tr> <tr> <td>103</td><td>Yana</td><td>Ahmedabad</td><td>12.21</td></tr> <tr> <td>104</td><td>Vashi</td><td>Kosamba</td><td>11.11</td></tr> </tbody> </table>	<u>snum</u> number(4) primary key	<u>sname</u> varchar2(20)	<u>city</u> varchar2(15)	<u>comm</u> number(5,2)	101	Devendra	Ahmedabad	4.5	102	Yogini	Ahmedabad	25.52	103	Yana	Ahmedabad	12.21	104	Vashi	Kosamba	11.11	3	1
<u>snum</u> number(4) primary key	<u>sname</u> varchar2(20)	<u>city</u> varchar2(15)	<u>comm</u> number(5,2)																				
101	Devendra	Ahmedabad	4.5																				
102	Yogini	Ahmedabad	25.52																				
103	Yana	Ahmedabad	12.21																				
104	Vashi	Kosamba	11.11																				

SQL> insert into **customer** values(&cnum,&cname','&city','&rating,&snum);

<u>cnum</u> number(4) primary key	<u>cname</u> varchar2(20)	<u>city</u> varchar2(15)	<u>rating</u> number(4)	<u>snum</u> number(4) references salespeople)
201	Rajanikant	Ahmedabad	1	101
202	Mohanlal	Boridara	2	102
203	Sanjay	Ankleshwar	3	102
204	Allu	Surat	4	101

SQL> insert into **orders** values(&onum,&amt,'&odate',&cnum, &snum);

<u>onum</u> number(4) primary key	<u>amt</u> number(6,2)	<u>odate</u> date	<u>cnum</u> number(4) references customer	<u>snum</u> number(4) references salespeople)
301	10000	'01-jan-2006'	201	101
302	20000	'15-feb-2010'	202	104
303	30000	'14-feb-2004'	203	103
304	40000	'15-feb-2003'	204	102

Write a query that produces the salesperson table with the columns in the following order: city, sname, snum, comm.

Write a SELECT command that produces the order number, amount, and date for all rows in the order table

Write a query that produces all rows from the customer table for which the salesperson's number is 101.

Write a query to display the only salesman no from the orders table.

Write a query that will give you all orders for more than Rs. 100/-.

Write a query that produces all customers serviced by salespeople who is located in Belgaum with a commission above 10%. Output the salespeople's name and city.

	Write a query on the Customers table whose output will exclude all customers with a rating <1; they are not located in Ahmedabad. Write a query that will produce all orders taken on October 14th Feb 2004 or 15th Feb 2005. Write a query to select cname, sname from customer and snum should be matched with both customer and salespeople table. Write a query to select cname, sname from customer and snum should be matched with both customer and salespeople table and sname should be from ('Devendra Vashi','Yogini Vashi') only. Write a query that selects all customers whose names begin with A or B. Write a query that will produce all of the customers whose names begin with a letter from A to G.		
2	Write and Perform the SQL queries for the following: SQL Where ===== 1. Select all records where the Ciry column has the value "Ahmedabad". 2. Use the NOT keyword to select all records where CiTy is NOT "Ahmedabad". 3. Select all records where the comm column has the value 11.11. 4. Select all records where the Ciry column has the value 'Ahmedabad' and the comm column has the value 25.52. 5. Select all records where the Ciry column has the value 'Ahmedabad' or 'Kosamba'. SQL Order By ===== 1. Select all records from the salespeople table and sort the results alphabetically by the column City. 2. Select all records from the salespeople table and sort the results reversed alphabetically by the column City. 3. Select all records from the salespeople table and sort the result alphabetically, first by the column CIRY, then by the column comm. SQL Null ===== 1. Select all records from the salespeople where the CITY column is empty. 2. Select all records from the salespeople where the salespeople where the CITY column is NOT empty.	3	2

	SQL Update ===== 1. Update the City column of all records in the salespeople table. 2. Set the value of the city columns to 'Surat', but only the ones where the Comm column has the value 88.99. 3. Update the salesperson name value and the City value for the particular salesperson. SQL Delete ===== 1.Delete all the records from the orders table where the onum value is 305. 2.Delete all the records from the orders table.		
3	Write and Perform the SQL queries for the following: 1. From the following table, create a view for those salespeople who belong to the city of New York. table: salesman 2. From the following table, create a view for all salespersons. Return salesperson ID, name, and city. table: salesman 3. From the following table, create a view to locate the salespeople in the city 'New York'. table: salesman 4. From the following table, create a view that counts the number of customers in each grade. Sample table: customer 5. From the following table, create a view to count the number of unique customers, compute the average and the total purchase amount of customer orders by each date. Sample table : orders 6. From the following tables, create a view to get the salesperson and customer by name. Return order name, purchase amount, salesperson ID, name, customer name. table: salesman, customer, orders 7. From the following table, create a view to find the salesperson who handles a customer who makes the highest order of the day. Return order date, salesperson ID, name. table: salesman, orders 8. From the following table, create a view to find the salesperson who deals with the customer with the highest order at least three times per day. Return salesperson ID and name. table: customer, salesman 9. From the following table, create a view to find all the customers who have the highest grade. Return all the fields of customer. table: customer 10. From the following table, create a view to count the number of salespeople in each city. Return city, number of salespersons. table: salesman 11. From the following table, create a view to compute the average purchase amount and total purchase amount for each salesperson. Return name, average purchase and total purchase amount. (Assume all names are unique.).	3	3

	Sample table: salesman, orders 12. From the following table, create a view to identify salespeople who work with multiple clients. Return all the fields of salesperson. Sample table: salesman, customer 13. From the following table, create a view that shows all matching customers with salespeople, ensuring that at least one customer in the city of the customer is served by the salesperson in the city of the salesperson. Sample table: salesman, customer 14. From the following table, create a view to display the number of orders per day. Return order date and number of orders. Sample table: orders 15. From the following table, create a view to find the salespeople who placed orders on October 10th, 2012. Return all the fields of salesperson. Sample table: salesman, orders 16. From the following table, create a view to find the salespersons who issued orders on either August 17th, 2012 or October 10th, 2012. Return salesperson ID, order number and customer ID. Sample table: orders		
4	Write and Perform the SQL queries to implement different operators(i.e. +,-,/,*,%) and SQL Aggregate Functions. (Assume the necessary database OR use the database of Practical-1:) Write a SQL statement to find the total purchase amount of all orders Write a SQL statement to find the average purchase amount of all orders. Write a SQL statement to find the number of salesmen currently listing for all their customers. Write a SQL statement to know how many customers have listed their names. Write a SQL statement to find the number of customers who get at least a rating for his/her performance. Write a SQL statement to know the maximum and minimum purchase amount of all the orders. Write a SQL statement that selects the highest grade for each of the cities of the customers. Write a SQL statement to find the highest purchase amount ordered by each customer with their ID and highest purchase amount. Write a SQL statement to find the highest purchase amount ordered by each customer on a particular date with their ID, order date, and highest purchase amount. Write a SQL statement to find the highest purchase amount on the date '01-JAN-06' for each salesman with their ID Write a SQL statement to find the highest purchase amount with their ID and order date, for only those customers who have the highest purchase amount in a day is more than 100	3	2

5	Write a PL SQL program that finds the minimum of two values. Here, the procedure takes two numbers using the IN mode and returns their minimum using the OUT parameters.	3	4
6	(A). Write a PL SQL program that demonstrates Declaring, Defining, and Invoking a Simple PL/SQL Function that computes and returns the maximum of two values. (B). Write a PL SQL program which calculates the factorial of a given number by calling itself recursively	3	4
7	Write a code in PL/SQL TO create a trigger that automatically updates a 'last_modified' timestamp whenever a row in a specific table is updated.	3	4
8	(A). Write a PL SQL program that demonstrates the concept of table-based records . (B). Write a PL SQL program that demonstrates the concept of cursor-based records .	3	4
9	Perform the DCL commands (grant, revoke): Create a student table with the following fields: StudentCode, Name, Dateofbirth, Course, RollNo, Batch, Result. Grant and revoke the following security constraints: 1. User Falgun can retrieve the entire table 2. User Sumit can insert and delete the entire table. 3. Every user can retrieve his/her record only. 4. User Nalayak can retrieve the entire table and update on Course and RollNo only 5. User Tasleem can retrieve over Name, StudentCode, and Result only. 6. User Wheel can retrieve as T and update as N 7. User Pratap can all the privileges for BE-IT student records. 8. User Jalak can delete records for students of Batch B2 9. GRANT ALL ON Student TO Jalak WHERE Course = 'BE-IT'; 10. User Babul can update and delete students record of courses where there are no more than 5 student 11. User Kanjus can retrieve the Eldest and youngest student	3	4
10	Joins on Tables: (considering the Practical-1 Tables perform the below joins) 1. Equi Join (List customer names along with their salesperson names (matching snum).) 2. Self Join (List pairs of salespeople who are in the same city but are not the same person.) 3. Natural Join (List order details along with customer information (joins automatically on same column names like cnum or snum).) 4. Theta Join (List orders where the order amount is greater than 20,000, along with salesperson name.) Use of Set operators: (considering the Practical-1 Tables perform the below set operators) 1. Union (Get a list of all cities where either a salesperson or a customer is located (duplicates removed).) 2. Intersect (Get a list of cities that are common to both salespeople and customers.) 3. Minus (Except) (Get a list of cities where salespeople are located but no customer is located.)	3	3

	Total	30	